

Executive Summary

The UK's marine fisheries present a mixed picture of sustainability. Official statistics report that UK vessels landed **719 kt** of fish in 2023 (up 12% on 2022) worth £1.1 billion [3†L38-L46] . By weight the catch is dominated by pelagic species (mackerel 106,533 t; herring 46,833 t) and demersals (haddock 38,397 t; Nephrops 27,402 t) [54†L150-L158] . By value, pelagics and shellfish dominate: mackerel (£132 M) and Nephrops (£111 M) are top, followed by scallops, cod and monkfish [54†L160-L167] . Key fishing grounds are in the North Sea and Scottish waters (Peterhead and Lerwick ports) for pelagics, and around Cornwall/Bristol Channel (Newlyn, Brixham) for groundfish [52†L823-L831] . **Stock assessments** indicate many UK stocks remain below sustainable levels. For example, North Sea cod biomass is far below targets and quotas were cut 20% for 2025 [61†L128-L137] . Haddock quotas were cut 5%, while plaice and saithe TACs were increased based on better biomass [61†L128-L137]

[61†L152-L160] . Crustacean stocks (lobster, crab) show high exploitation: e.g. Northumberland lobster have $F > F_{MSY}$ and female biomass near limit reference [24†L440-L449] ; similarly Central North Sea edible crab has fishing mortality above F_{MSY} and male stock at Blim [27†L353-L355] . Scallop stocks are variable: Eastern Channel and Lyme Bay areas are relatively productive (Lyme Bay harvest rate above MSY), whereas Western Channel offshore has seen large biomass declines [30†L984-L993] [30†L1000-L1004] . Many small demersals and cephalopods are data-poor (ICES Category 3, with only provisional MSY proxies) [8†L349-L358] .

Spatially, fishing activity clusters on traditional grounds (North Sea, Celtic Sea, Channel) and many UK Marine Protected Areas overlap these grounds. The UK now has **377 MPAs covering ~38% of its waters** [59†L45-L48] . Recent byelaws protect sites like Dogger Bank and deep-sea canyons from bottom trawling [59†L75-L83] . Despite these protections, Greenpeace analysis found >1.3 Mt of fish (2020–24) have still been caught inside MPAs, highlighting enforcement gaps [59†L45-L48] .

Climate change is altering species distributions. A 2025 Cefas study found ~70% of UK fish species have already shifted northwards in response to warming [33†L51-L60] . “Winners” such as rays (spurdog, thornback) and native oyster are gaining habitat in UK waters, while sessile invertebrates (sea pens, fan mussels) face up to 40% habitat loss by century's end [33†L51-L60] . Mobile species may re-colonise new areas, but many over-wintering spawning grounds (e.g. southern North Sea) are threatened by warming [33†L77-L85] . Invasive non-native species are also a concern: species like Pacific oyster and Chinese mitten crab can alter habitats and fisheries. Scottish reports list widespread invaders (Codium algae, slipper limpets, etc) that smother habitat, with an estimated £246 M/yr economic impact in Scotland alone [35†L72-L82] [35†L148-L153] .

Economically, the UK fleet had turnover ~£1.075 B in 2023 (profit £256 M) [39†L47-L55] . About 24% of landing value is taken abroad by UK vessels. Scotland lands the majority of volume and value, reflecting its large pelagic fleet [39†L47-L55] . UK vessels export ~£1.7 B of seafood (2023), mainly whitefish and pelagics: 70% to the EU (France £779 M alone) and growing trade to Asia, despite Brexit hurdles [57†L19-L24] [57†L27-L32] . The **supply chain** is highly globalized (flowchart below), with most UK landings auctioned domestically but many consignments processed and sold overseas, especially cod/haddock into EU markets.

Data gaps and uncertainties remain substantial: many coastal species lack surveys, bycatch (esp. of skates, sharks) is poorly quantified, and the contribution of un-dredged scallop beds is unknown (likely

leading to overestimate of harvest rates) [30†L1000-L1004] . Stock assessments use survey proxies or rough MSY buffers for data-poor stocks [8†L349-L358] . The quality of stock and effort data is uneven, especially for small vessels and non-quotas.

Policy implications: UK policy (Fisheries Act 2020) mandates precautionary, scientific-based quotas and an ecosystem approach. In practice, UK quotas are negotiated trilaterally (with EU/Norway/Faroe), aiming to meet ICES advice [61†L107-L115] , but socio-economic factors have dampened cuts (e.g. haddock cut only 5% despite low biomass). The growing MPA network and stricter byelaws reflect a shift to habitat protection, but enforcement and balancing with fishing rights is contentious. Climate-driven shifts underscore the need for adaptive management and transboundary cooperation to allow dynamic stock ranges. Major gaps in data and monitoring must be addressed to reduce uncertainty in status estimates. Overall, many UK fisheries remain on the cusp of sustainability: continued reductions in fishing mortality (where over $F < MSY$) and habitat protections are recommended to rebuild stocks to safe biomass levels. Support for coastal communities (jobs and processing) should accompany such measures to ensure long-term economic and environmental viability.

Table 1. Selected UK fisheries: stock status and management (2023). For each species-stock, biomass (B) and fishing mortality (F) are compared to MSY reference points [61†L128-L137] [24†L440-L449] [27†L353-L355] [30†L984-L993] . For simplicity, “~MSY” means near target, “<MSY” means below biomass target (overfished), “>MSY” means biomass above target; F status is “>1” if $F > MSY$ (overexploited), “~1” if $F \approx MSY$. Management column notes main measures (TAC = total allowable catch/quota; MPA restrictions; technical measures).

Species (stock)	B vs $B < MSY$	F vs $F < MSY$	Status	Management
Cod (North Sea)	<<1 (far below)	>1	Overfished, F too high	TAC cut 20% (2025); bycatch regulations [61†L128-L137]
Haddock (North Sea)	<1 (below)	≈ 1	Rebuilding; near target	TAC -5% (2025) [61†L128-L137]
Herring (N. Sea)	≈ 1	<1	Stable; healthy biomass	TAC -24% (2025) [61†L128-L137] (restored)
Plaice (N. Sea)	>1	<1	Above target (good)	TAC +14% (2025) [61†L152-L160]
Saithe (N. Sea)	≈ 1	<1	Stable	TAC +7% (2025) [61†L158-L163]
Whiting (N. Sea)	<1 (below)	<1	Rebuilding	TAC +46% (2025) [61†L164-L170]
Mackerel (North Sea)	>1	<1	Healthy	TAC (share)†; UK ~53800 t (2025)
Nephrops (N. Sea)	≈ 1 (varies by area)	≈ 1	Generally sustainable	TAC/quota per bed; FPs (fishing plans)

Species (stock)	B vs B_{MSY}	F vs F_{MSY}	Status	Management
Lobster (England E-Coast)	≈1 (females); <1 (males)	>1 (esp. males)	Marginal: F > target	Minimum size, v-notching; no TAC (non-quota species) 【24†L455-L463】
Edible crab (CNS)	≈1 (females); <1 (males)	>1 (males)	Stock near limit	Crustacean permit area; no TAL (non-quota) 【27†L353-L355】
King scallop (English Channel)	<1 (some beds declining)	>1 in places	Mixed: some overexploited (Lyme Bay) 【30†L992-L1000】	No EU quota; project-based management, some closures; FSA (dredge surveys)
Spurdog (S. Dogfish, Atlantic)	>1 (rebuilding)	≈0.6 (below target)	Recovered (precaution) 【64†L284-L293】	Zero-TAC (2010–22), now precautionary TAC reinstated 【64†L284-L293】
Common skate (spp.)	<1 (below)	<1	Critically depleted	Protected (EU prohibited)

Sources: ICES/Cefas/Defra stock reports and advice 【61†L128-L137】 【24†L440-L449】 【27†L353-L355】 【30†L984-L993】 ; Cefas (2023–24).

Table 2. Top UK landings by species (2023). Landings (all vessels, domestic & foreign) and first-hand value for top species, from MMO/Seafish data 【54†L150-L158】 【54†L160-L167】 . Landings volumes are dominated by pelagic fish, whereas value is driven by high-value shellfish and whitefish. “–” indicates data not reported here.

Species	Landings (2023, tonnes) 【54†L150-L158】	First-hand value (2023, £M) 【54†L160-L167】
Mackerel	106,533	132
Nephrops	27,402	111
King scallop	–	60
Cod	–	45.2
Monkfish	–	44.9

Methods

This report synthesizes data from primary official sources and recent scientific literature. We reviewed UK government statistics (Defra/MMO, Cefas, devolved administrations) and industry data (Seafish) for

catch volumes, values, and fleet economics. Stock status is based on ICES and Cefas assessment reports (2022–2024) and Fisheries Management Plans. Spatial patterns are inferred from fisheries statistics (e.g. landings by port [52†L823-L831]) and known stock ranges. We incorporate published biodiversity datasets (JNCC/Cefas, IUCN Red List) and recent climate-impact studies [33†L51-L60] . Key findings are tabulated and illustrated (tables and mermaid flowchart) using cited data. Data gaps (e.g. partial survey coverage) are noted qualitatively, per source documentation [8†L349-L358] [30†L1000-L1004] .

Results

Stock status and management

Many commercially important UK stocks are not at sustainable levels. For example, North Sea cod biomass remains well below B_{MSY} and fishing mortality above F_{MSY} , prompting a TAC cut of 20% for 2025 [61†L128-L137] . Haddock is closer to target but still not fully rebuilt, with a 5% TAC reduction [61†L128-L137] . Plaice and saithe are in better shape (biomass above targets), allowing TAC increases [61†L152-L160] [61†L158-L163] . Pelagic stocks (mackerel, herring) are in good condition; ICES advises sustainable or increased catches, reflected in negotiated TACs (~£310 M UK share in 290 kt NS stocks [61†L86-L94]). Nephrops populations are generally stable, though individual fishing grounds vary and are managed via effort limits/TACs per bed. Non-quota species (e.g. many rays, cephalopods, and small groundfish) lack TACs and rely on byelaws or technical measures.

Selective management measures are in place. The UK negotiates annual TACs with the EU/Norway; UK shares for North Sea 2025 include 9,419 t cod and 62,128 t haddock [61†L128-L137] . The **Fisheries Act 2020** mandates setting quotas consistent with scientific advice. In practice, socioeconomic considerations have moderated some cuts (e.g. only –5% haddock TAC despite low stock). Technical measures (minimum sizes, net restrictions) apply to many stocks. The growing MPA network (377 sites, 38% coverage [59†L45-L48]) has introduced gear restrictions: in several MPAs bottom trawls and dredges are banned [59†L75-L83] . Three Highly Protected Marine Areas were designated in 2023 where all fishing is prohibited. Combined, these measures aim to reduce pressure in critical habitats (Dogger Bank, offshore reefs, estuaries).

Spatial distribution of fisheries

Fishing activity is concentrated in well-known grounds. Scottish vessels (pelagic fleet) mostly fish the northern and western North Sea (Shetland, Hebrides) – reflected by high landings at Peterhead and Lerwick [52†L823-L831] . South-west English ports (Newlyn, Brixham) specialize in groundfish and shellfish from the Celtic Sea/English Channel [52†L823-L831] . Spatially explicit data (e.g. VMS/landings maps) show intense trawling in the North Sea and mid-Channel, with static gear (pots, nets) in inshore waters. Our report lacked uniform GIS data for species distribution mapping, but published “fishery sensitivity” maps and survey grids indicate main spawning/nursery areas for cod, haddock, plaice, etc., aligning with the fishing grounds above (e.g. Dogger Bank for sand eels/cod). [8†L349-L358]

flowchart LR

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A[Fishing Catch at Sea] --> B[Port Landing/Auction]
B --> C[Processing \& Distribution]
C --> D[Wholesale \& Retail Markets]
D --> E[Consumers (Domestic & Export)]

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Figure: Simplified UK seafood supply chain. Fish caught by UK vessels are landed to auction ports and local processors; most high-value catch (e.g. cod, shellfish) is then processed and largely exported abroad, while some is sold in domestic markets (e.g. mackerel and herring).

Economic value and trade

UK landings rose in 2023, reversing a multi-year decline [3†L38-L46] . Pelagics and shellfish saw value increases, while demersal whitefish value fell slightly. Overall turnover of the catching sector was £1.075 billion (2023) [39†L47-L55] . The UK is a net exporter of fish: exports totaled ~£1.7 billion in 2023, up 1.3% from 2022 [57†L19-L24] . France is the largest market (£779 M, 43% of exports) [57†L27-L32] ; most UK exports are to the EU (70% by value) [57†L19-L24] . Imports (~£3.5 B) come mainly from outside the EU (76%), reflecting demand for whitefish and tropical species not produced domestically [57†L19-L24] .

Table 2 (above) summarizes landings volumes and first-sale values for key species. Mackerel (106,533 t) and Nephrops (27,402 t) lead in volume [54†L150-L158] , while the highest values come from mackerel (£132 M) and Nephrops (£111 M) [54†L160-L167] . King scallop and monkfish, though smaller in volume, contribute significant value. Prices (first-hand) have been relatively stable; wholesale/retail prices are higher after processing and marketing. There is no single UK ex-vessel “price index,” but aggregated industry data suggest real-term landings prices have roughly tracked inflation over the last decade.

Species diversity and conservation status

UK seas host hundreds of fish and invertebrate species. Most commercial catches come from a few dozen species, but true biodiversity is higher when including bycatch and non-targets. Marine Protected Area network data show many habitats and species formally protected. However, **IUCN Red List** conservation statuses highlight concerns for certain taxa. Critically endangered species in UK waters include common skate (†*Dipturus batis*) and spurdog (*Squalus acanthias*) [64†L284-L293] . Spurdog (once IUCN-CR in NE Atlantic) has been protected (zero-TAC since 2010) and is now rebuilding [64†L284-L293] . Other sharks/rays (undulate, manta, lanternshark) are vulnerable or data-deficient; coastal fish like salmon and sturgeon (extirpated) also figure on Red Lists. Invertebrates such as native oyster and seagrass have suffered >95% declines; native oyster has very limited reproduction and may need reseedling programs (though climate models paradoxically show doubled suitable habitat area by 2100 [33†L59-L64]).

Species richness varies regionally: warmer southern waters have historically contained more thermophilic species (bass, hake) than the cooler north. Climate-driven range shifts are already changing the mix. Long-term plankton and fish surveys (e.g. UK-wide Fish Atlases) document new arrivals (e.g. red mullet, rays) in the Channel and Irish Sea. Overall marine biodiversity indicators (e.g. MSFD Biodiversity Status) are improving slowly under MPAs, but the full metric of “species richness” is not explicitly tracked by UK agencies.

Invasive species

Non-native species threaten UK marine ecosystems. Introduced via shipping or aquaculture, invaders like the American lobster (*Homarus americanus*) and Pacific oyster (*Crassostrea gigas*) can compete with natives or alter habitats. England and Scotland report numerous marine invasives: invasive seaweeds (*Sargassum muticum*, *Codium*), Bryozoa (*Schizoporella*), ascidians (*Styela*, *Didemnum*) and invertebrates (*Austrominius* barnacles) are widespread in UK waters [35†L72-L82] . These can foul

gear and infrastructure, reduce native stocks, and spread diseases. Early detection efforts are in place but removal is difficult once established [35†L49-L57] [35†L72-L82] . Economic costs are significant: one study estimated invasive non-natives cost Scottish marine industries £246 M/year [35†L148-L153] .

Climate change impacts

Warming seas are already shifting UK fish distributions. Cefas and MCCIP studies indicate northern range expansions for ~70% of species. Species currently scarce or absent in northern waters (e.g. seafans, deep corals) are expected to lose up to 40% habitat suitability, while warm-water species (bass, bream) may increase in the North Sea [33†L51-L60] . This may initially boost some fisheries but risks ecosystem mismatches. Southern hotspots like the south North Sea and western Channel warm fastest, threatening local stocks. The *Cefas* modelling stresses creating “climate corridors” (through protected area design) to let species move, and adjusting stock units if biological populations shift offshore or poleward [33†L77-L85] . Ocean acidification and deoxygenation (not detailed here) will further stress shellfish and reef habitats, potentially impacting groundfish indirectly.

Data gaps and uncertainties

Data limitations constrain management. Many stocks (e.g. rays, coastal cod, squid species) lack scientific surveys; assessments rely on catch indices or cross-jurisdictional data. Bycatch (particularly of protected elasmobranchs) is poorly quantified. Some assessment methods (e.g. for Nephrops) assume constant catchability or use broad MSY proxies for Category 3 stocks [8†L349-L358] . For king scallops, current assessments cover only dredged beds; unfished grounds (which may contribute larvae) are omitted [30†L1000-L1004] , biasing harvest estimates. Socio-economic data (crew numbers, sector profitability) are better, but still sample-based. Finally, post-Brexit regulatory changes and UK/EU trade frictions add uncertainty to future landings/trade, not yet fully reflected in data.

Discussion

Overall, UK fisheries management has moved towards science-based quotas and habitat protection, but many stocks remain exploited at marginal levels. Several key species have shown recovery (haddock, plaice, mackerel), suggesting management can work when followed. Others (cod, rays, shellfish) require sustained cuts or protected areas to rebuild. The economic value chain is polarized: high-value shellfish support coastal economies, whereas bulk pelagics, though large-volume, have lower unit value. The supply-chain flowchart (above) underscores the importance of export markets; policy must consider market access (trade deals) in parallel with stock rules. Climate change and invasive species compound challenges by shifting baselines and introducing new competitors/predators.

The tables illustrate the uneven status: e.g. North Sea cod is still “in crisis” despite years of cuts, while some scallop beds have transient gluts. Management must be adaptive and precautionary. Brexit (Fisheries Act) gives the UK more control, but also more responsibility for data collection and enforcement, especially in inshore (0–6 nm) zones where devolved governments manage fisheries. Harmonizing approaches across nations (UK, EU, Norway) is essential, as fish do not respect boundaries. Protecting juveniles and critical habitats via MPAs (e.g. Ice-age reefs, seagrass beds) should complement quotas.

Data gaps need urgent attention. Expanding fishery-independent surveys (e.g. multi-species trawl surveys, acoustic monitoring) would fill knowledge holes. Improved catch reporting (including recreational fishing, small boats, discards) would refine stock status. Routine reassessment of stock

units under shifting conditions is needed: TACs and quotas should be revisited if fish migrations alter stock composition.

Recommendations

- **Strengthen science-based quotas:** Implement ICES advice fully, reducing catches where $F > F_{MSY}$ and increasing where stocks rebound. Consider multi-year management plans with automatic rebuilding triggers.
- **Expand monitoring:** Invest in fish and shellfish surveys (including new technologies like eDNA, drones, sonar) especially for data-poor species. Mandate better logbook coverage of bycatch.
- **Integrate climate considerations:** Review TAC zones and stock boundaries in light of warming (northward shifts), and incorporate climate risk into the MPA network (protect future “hotspots”). Use fisheries-management plans to require “climate screening” of measures.
- **Enhance MPA enforcement and networks:** Ensure MPA byelaws (e.g. trawl bans) are enforced. Consider adding high-value spawning grounds (e.g. sandeel banks) to protected areas. Develop management plans for HPAs.
- **Address invasive species:** Improve biosecurity (hull cleaning, ballast regulation) to prevent new arrivals. Continue eradication/containment efforts for known threats (e.g. remove Chinese mitten crabs where feasible).
- **Support supply chain resilience:** Provide technical and financial support for domestic processing to reduce reliance on volatile export markets. Monitor economic indicators (fleet profitability, price changes) to adjust quota trade-offs.
- **Coordinate policy across sectors:** Fisheries policy should align with biodiversity (Biodiversity Net Gain commitments) and climate goals. For example, byelaws in MPAs also benefit carbon-rich seabeds. Cross-departmental oversight (Defra, BEIS, etc.) can ensure consistency.

Datasets & Sources: Key data sources include the UK Sea Fisheries Statistics (MMO/Defra), Cefas stock status reports (2022–24), ICES advice and databases, Seafish economics reports, and research publications (e.g. Cefas 2025 climate study [33†L51-L60]). Table 1 uses data from ICES/Defra stock assessments [61†L128-L137] [24†L440-L449] [27†L353-L355] [30†L984-L993] . Table 2 uses MMO/Seafish landing/value data [54†L150-L158] [54†L160-L167] . Spatial information is based on government landings-by-port data [52†L823-L831] and MPA network statistics [59†L45-L48] . All sources are cited inline above.
